06898

Active case detection and treatment of malaria in pregnancy using LAMP technology (LAMPREG): A pragmatic randomised diagnostic outcomes trial - interim results

Late Breakers

22. Clinical trials
Likely attendance
Onsite

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Background

According to the WHO, in 38 countries in Africa there were an estimated 40 million pregnancies of which 13.3 million (32%) were exposed to malaria infection during pregnancy (MiP). MiP leads to low birthweight (LBW), a strong risk factor for childhood mortality. Intermittent preventive therapy in pregnancy (IPTP) can reduce MiP but uptake in Africa remains low. This study aims to evaluate active detection of MiP using a molecular method by loop-mediated isothermal amplification (LAMP) during antenatal care (ANC) in Ethiopia where there is no IPTP.

Methods

A pragmatic randomized diagnostic outcomes trial is being conducted at three rural hospitals and five health centres in moderate to high transmission regions of Ethiopia. Women are randomized to either the standard of care (SOC, 1/3) or the active case detection arm (LAMP, 2/3) at their first ANC visit and subsequently followed through to delivery. Malaria diagnosis is performed by microscopy and RDT in the SOC arm using Malaria-LAMP (LoopampTM, Human Diagnostica, Wiesbaden, Germany), microscopy, and RDT in the intervention arm. Treatment of women positive for malaria by any method was according to the Ethiopian national guidelines.

Results

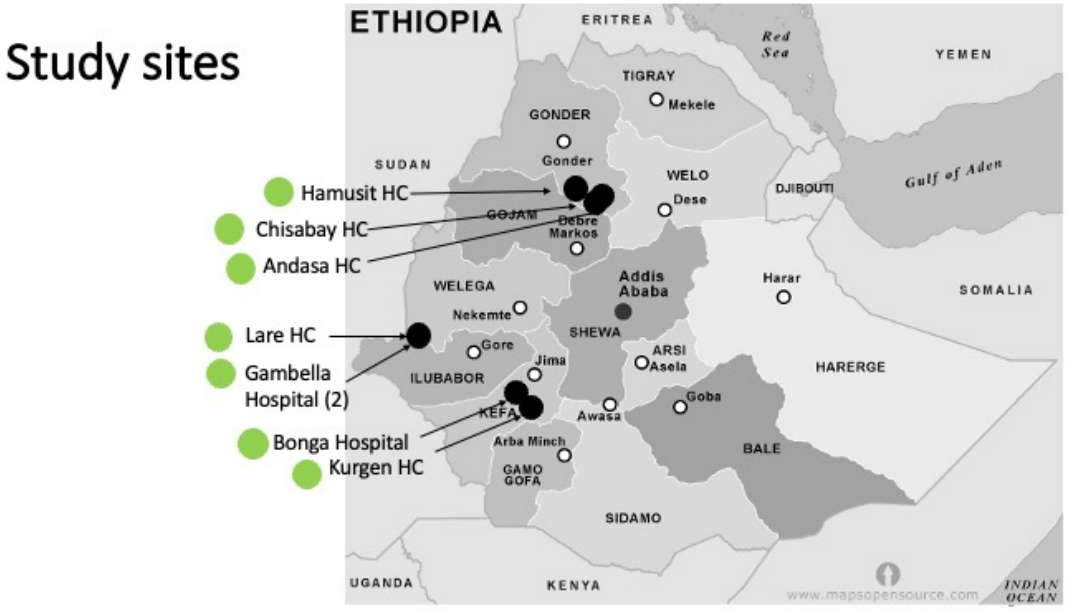
In the interim analysis, 2,116 women were enrolled (715 SOC and 140 LAMP) with 1452 deliveries completed. At inclusion, there were no significant differences in terms of maternal age (p=0.55), hemoglobin (p=0.64), or gravidity (p=0.25). Malaria detection was superior using LAMP (219 positive, 15.6%), compared to microscopy (9.2%) and RDT (8.1%). At delivery, birthweight was 47.0g higher in the LAMP arm (p=0.12). The proportion of LBW

was 5.06% (SOC) and 4.24% (LAMP) (p=0.57). Newborn anemia was 3.85% (SOC) versus 2.31% (LAMP) (p=0.13). Premature deliveries were 12.12% (SOC) versus 5.26% (LAMP) (p=<0.0001) and 28-day newborn weight 4,121g (SOC) versus 4,331g (LAMP) (p=<0.0001).

Conclusions

Molecular diagnosis of malaria with LAMP detected double the number of MiP compared to standard microscopy and RDT in this randomized trial. Active case detection of MiP using LAMP technology resulted in improved birth and 28-day weight, lower LBW, less newborn anemia and prematurity. Molecular diagnosis of malaria is feasible to implement in antenatal clinics in sub-Saharan Africa with demonstrable improvements in critical MiP outcomes.

Study sites in Ethiopia



Diagnostic Performance

Diagnostic Method	SOC arm (n=715)	LAMP arm (n=1401)
Microscopy	31	129
RDT	31	114
LAMP	N/A	219

Outcome Measures

Outcome Measure	SOC arm (n=715)	LAMP arm (n=1401)	*p*
maternal age (enrolment) *(mean, STDev)*	26.01 [4.57]	26.13 [4.57]	0.55
Maternal Hemoglobin (enrolment), g/dL *(mean, STDev)* Anemia (Hb < 11g/dL [%])	12.52 [1.548] 5.87%	12.47 [1.55] 8.05%	0.64
Multigravid (enrolment), no.	575	957	0..25
Primigravid (enrolment), no.	266	495	0.25
Gestational age (enrolment), weeks *(mean, STDev)*	19.017 [3.584]	19.886 [3.514]	<0.0001
Low birthweight (%) (<2500g)	5.06%	4.24%	0.57
Preterm delivery *(%)*	12.12%	5.26%	<0.0001
Newborn weight day-1, g *(mean, STDev)*	3184 [445]	3230.57 [498]	0.12
Newborn weight day-28, g (mean, STDev)	4121.27 [672.85]	4330.77 [837.29]	<0.0001
Newborn hemoglobin (enrolment), g/dL *(mean, STDev)* Anemia (Hb < 11g/dL [%])	18.07 [2.87] 3.85%	18.19 [2.76] 2.31%	0.13

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Conflicts of interest

Do you have any conflicts of interest to declare?
No